

Personalized Transformer for Explainable Recommendation

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Venue: ACL 2021



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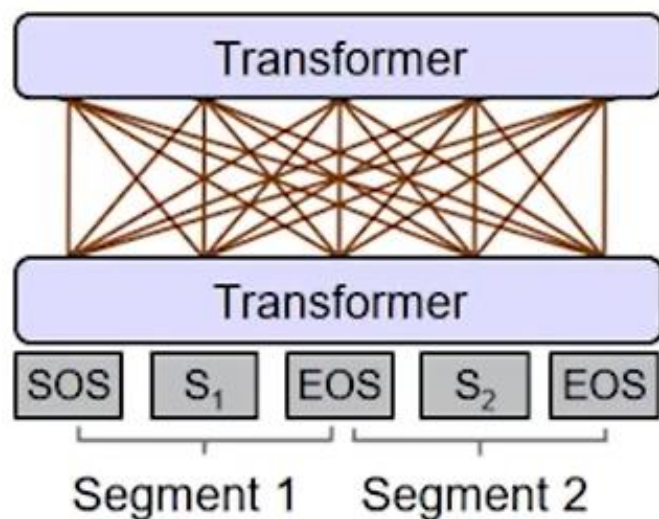
Course: Technical English Course

Instructor: Dr. Mataji



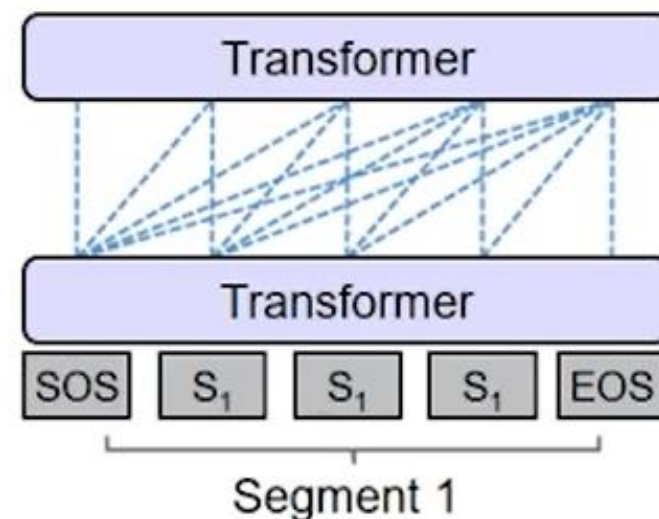
Transformer (Vaswani et al., NIPS'17)

BERT (Devlin et al., NAACL'19)



Bidirectional model for classification

OpenAI GPT (Radford et al., 18)



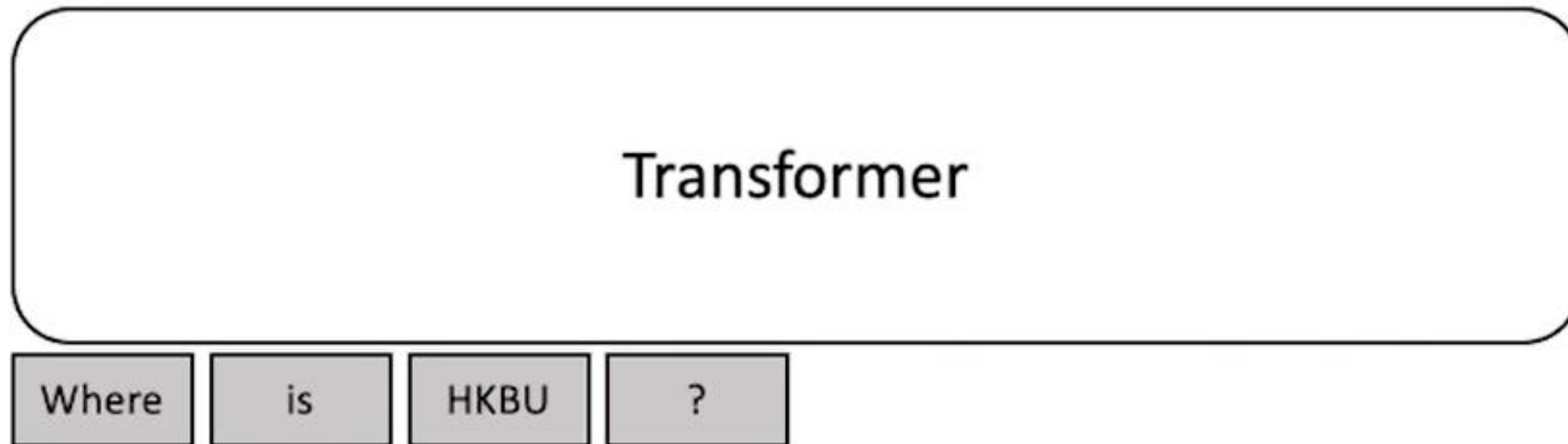
Unidirectional model for generation

VS.



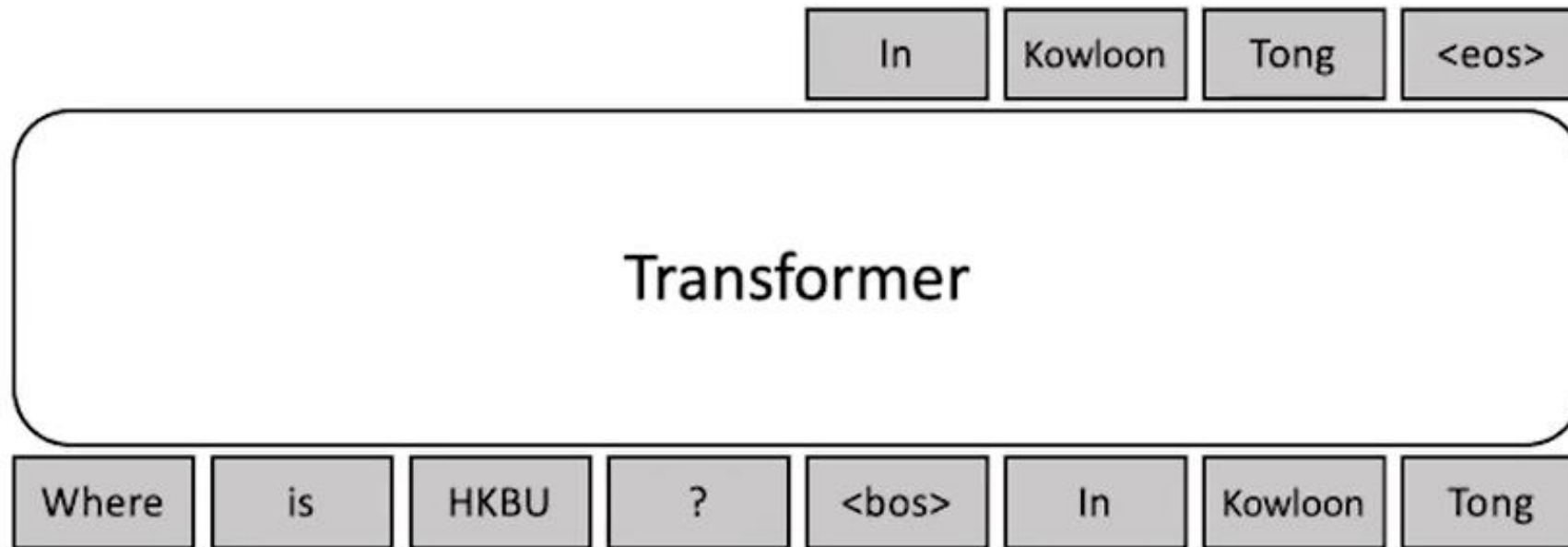
Autoregressive Natural Language Generation

- Predict future tokens based on past tokens
 - Generate an output sequence, based on the given input sequence



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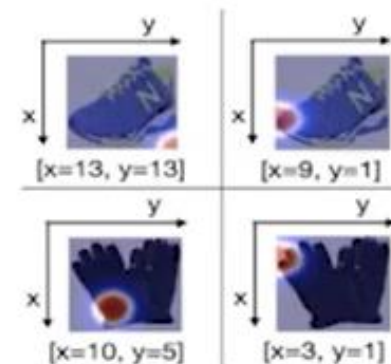


Explainable Recommendation

- Given a user-item pair, provide an **explanation** to justify why the item is recommended to the user
 - Pre-defined template (Zhang et al., SIGIR'14)
 - Image visualization (Chen et al., SIGIR'19)
 - Natural language sentence in this work
 - E.g., *"the style of the jacket is fashionable"*
 -

You might be interested in [feature],
on which this product performs well.

You might be interested in [feature],
on which this product performs poorly.



Transformer for Explanation Generation

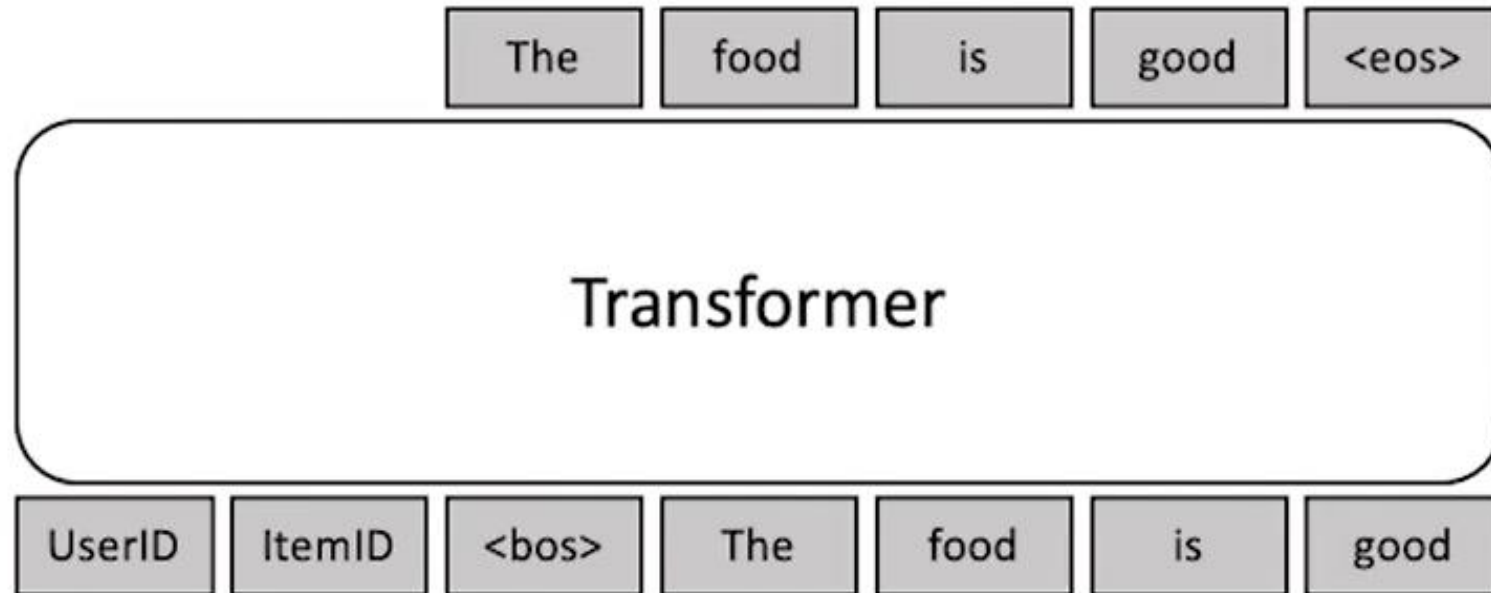
- Consider the user-item pair as an input sequence
 - Regard IDs as tokens, similar to words

Transformer



Transformer for Explanation Generation

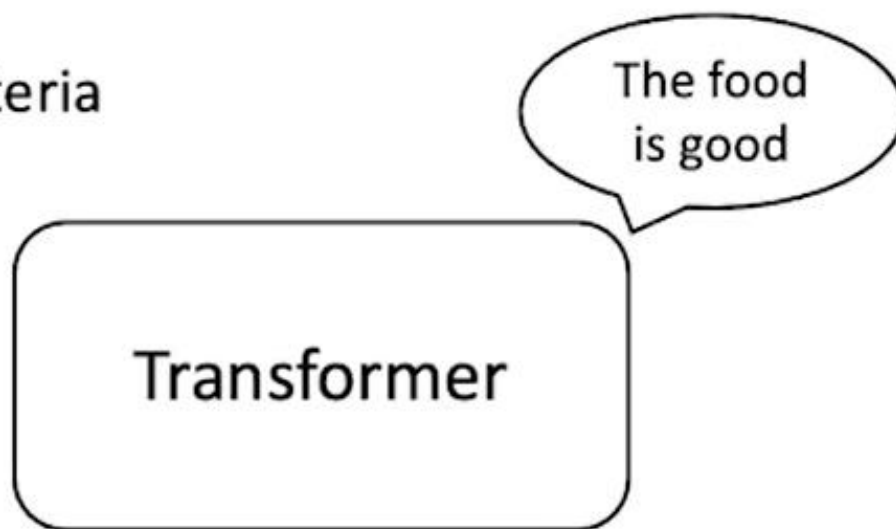
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Problem Identification

- Identical generated explanations for almost every user-item pair

- Adam Main Canteen
- Beth Renfrew Cafeteria
- Carol Bistro Bon
- David Harmony Cafeteria
-



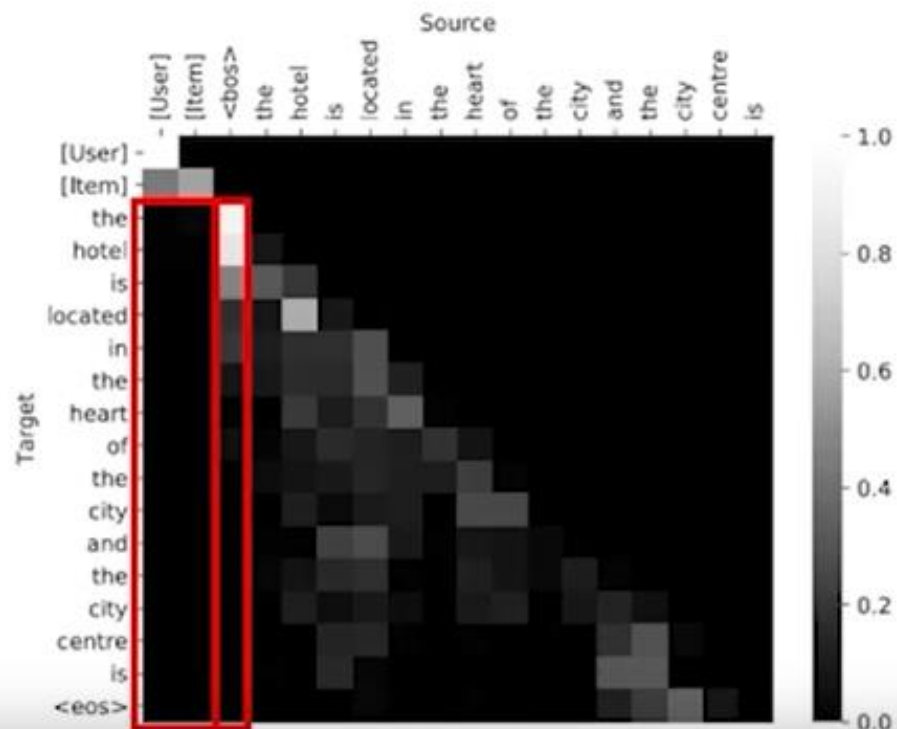
- Less useful, if unable to explain the key specialty of each recommendation
- Could cause negative effects on users (Tintarev and Mashoff, 15)



Attention Visualization

- The generation relies heavily on <bos>
 - The reason why all explanations are identical
- Attention weights on userID and itemID approach 0
 - Model insensitive to IDs

Why insensitive?



Problem Analysis

- Frequency mismatch between IDs and words
 - One user/item ID vs. hundreds of words in a review
 - An ID appears in only a few reviews
- IDs being regarded as uncommon words (OOV tokens)



Shelby S. Elite '2021
Saint Paul, MN
📍 166 📷 354 📝 771

★★★★★ 12/4/2019

📷 6 photos

Ho Lee Fook was one of the best food spots I went to in HK. At first I was skeptical because sometimes the fusion or westernized type Asian restaurants are all for the look but don't taste great. But, Ho Lee Fook was beautiful inside and the food was amazing. We ordered the pan fried thick rolled noodles and the massive bone steak (forgot the actually name) but you won't miss it on the menu. The noodles were crispy and seasoned just right. The steak was so tender and delicious. It came with a jalapeño sauce on the plate which complimented it so well.

While being here I forgot I was in HK because everyone spoke English and the menu was also in English! The entrance is so cute with the lucky cats all on the walls.

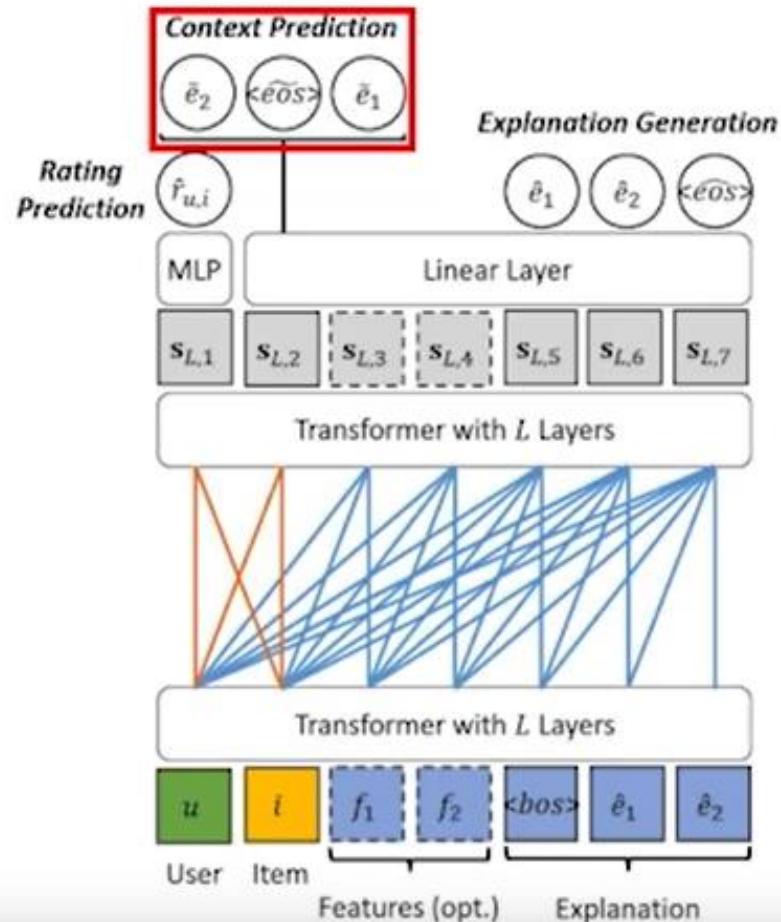
If you are visiting HK or live there I definitely recommend giving this place a try! It is a little on the pricey side but for the atmosphere it is expected.

Restaurant review
([yelp.com](https://www.yelp.com))



Solution: Context Prediction

- Bridge IDs and words, and give the former linguistic meanings



Context Prediction & Explanation Generation

- Context prediction: predict explanation words in one step
 - With the item representation

- Explanation generation: generate them one by one

- Linear layer

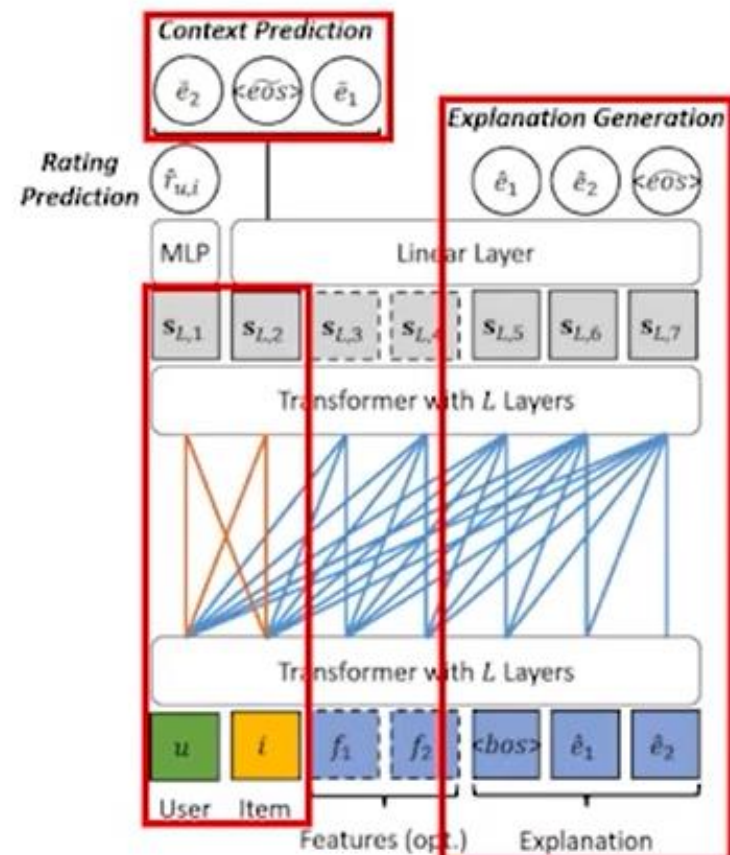
$$c_t = \text{softmax}(\mathbf{W}^v s_{L,t} + \mathbf{b}^v)$$

- NLL loss for context prediction

$$\mathcal{L}_c = \frac{1}{|\mathcal{T}|} \sum_{(u,i) \in \mathcal{T}} \frac{1}{|E_{u,i}|} \sum_{t=1}^{|E_{u,i}|} -\log c_2^{e_t}$$

- NLL loss for explanation generation

$$\mathcal{L}_e = \frac{1}{|\mathcal{T}|} \sum_{(u,i) \in \mathcal{T}} \frac{1}{|E_{u,i}|} \sum_{t=1}^{|E_{u,i}|} -\log c_{2+|F_{u,i}|+t}^{e_t}$$



Multi-task Learning

- Three tasks trained in an end-to-end manner
 - Explanation generation
 - Context prediction
 - Rating prediction

$$\mathcal{J} = \min_{\Theta} (\lambda_e \mathcal{L}_e + \lambda_c \mathcal{L}_c + \lambda_r \mathcal{L}_r)$$



Datasets (Li et al., CIKM'20)

- Yelp
 - Restaurant
- Amazon
 - Movies & TV
- TripAdvisor
 - Hotel
- The explanation is a review sentence containing features



	Yelp	Amazon	TripAdvisor
#users	27,147	7,506	9,765
#items	20,266	7,360	6,280
#records	1,293,247	441,783	320,023
#features	7,340	5,399	5,069
#records / user	47.64	58.86	32.77
#records / item	63.81	60.02	50.96
#words / exp	12.32	14.14	13.01

* **exp** denotes **explanation**.



Evaluation Metrics

- Recommendation
 - RMSE & MAE
- Explanation
 - Text quality: BLEU (Papineni et al., ACL'02) & ROUGE (Lin, ACL'04 Workshop)
 - Not equal to explainability (Chen et al., SIGIR'19 Workshop; Li et al., CIKM'20)
 - Explainability from the angle of item features (Li et al., CIKM'20)
 - Unique Sentence Ratio (USR)
 - Feature Matching Ratio (FMR)
 - Feature Coverage Ratio (FCR)
 - Feature Diversity (DIV)



Conclusion

- The 1st to enable Transformer with personalized natural language generation
 - Shed light on a broader scope of fields that also need personalization
 - E.g., personalized conversational systems
- Model small and unpretrained, but effective and efficient
 - Open up a new way of exploiting Transformer by designing good tasks instead of scaling up model size
- Design a task to connect IDs and words
 - Point out a way for Transformer to deal with heterogeneous inputs
 - E.g., **image generation** based on text in multimodal AI



Q&A

Thank you!



lileipisces.github.io

